Luo Zhou

Curriculum Vitae
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Ph.D. Candidate, Donghua University
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Personal Information

Native Place: Huaian, Jiangsu, China
Date of Birth: November 28, 1994

Research and Work Experience

- **IEEE International Conference on Distributed Computing Systems (ICDCS 2025)** *Jul 20 – Jul 23, 2025*
  Academic Exchange & Oral Presentation Glasgow, UK
  - Delivered an in-person oral presentation and discussed the technical details and experimental results with attendees.
  - Engaged in research exchanges across sessions and posters, collecting actionable feedback on problem formulation, evaluation design, and positioning.
  - Expanded my research network in mobile systems and sensing/security-related topics, informing subsequent iterations of my ongoing projects.
- **University of Electronic Science and Technology of China** *May 2023 – Oct 2023*
  Visiting Scholar Chengdu, China
  - Completed the development of an earable (earphone) sensing prototype during the research visit, laying a solid foundation for my subsequent Ph.D. research in wearable sensing and security.
  - Gained hands-on experience in hardware prototyping and iterative debugging, bridging sensing principles with end-to-end system realization.
  - Worked closely with the host group in a collaborative research setting, improving teamwork skills such as task coordination, progress tracking, and technical communication.
- **ACM Turing Award Celebration Conference (TURC 2023)** *Jul 28 – Jul 30, 2023*
  Academic Participant Wuhan, China
  - Attended invited talks and technical sessions, with a focus on mobile computing, AI systems, and security-related themes.
  - Participated in community activities and discussions with researchers and students, broadening my perspective on impactful research directions.
  - Reflected conference takeaways into my research planning, including reading lists and candidate problems for follow-up exploration.
- **Nanjing FiberHome Starry Sky Communication Co., Ltd.** *Jul 2020 – Jul 2022*
  Mobile & IoT Systems Engineer Nanjing, China
  - Worked in a carrier-grade product environment, gaining first-hand experience with real-world constraints in mobile/IoT systems (reliability, latency, scalability, maintainability).
  - Owned key service modules across the full lifecycle (architecture discussion, interface specification, data management, rollout and iteration), strengthening my ability to build end-to-end research systems.
  - Coordinated with cross-functional teams on production requirements and system integration, which aligns with my current research on deployable, privacy-preserving mobile/wearable computing.
- **China Aviation Industry Information Center** *Aug 2019 – May 2020*
  Industrial IoT Systems Engineer Nanjing, China
  - Maintained and extended ERP/MES systems in an aerospace-manufacturing setting, gaining exposure to industrial data flows and operational constraints typical of Industrial IoT.
  - Participated in system hardening and long-term maintenance (compatibility, reliability, traceability), building engineering discipline useful for reproducible security/sensing research deployments.
- **School-enterprise Collaboration Project** *May 2019 – Aug 2019*
  IoT Systems Project Engineer Zhenjiang, China
  - Delivered an end-to-end intelligent fire-alarm solution (mobile + backend), providing practical experience with IoT-style sensing/alert pipelines and real-time event handling.
  - Covered requirement analysis and data modeling, which supports my current research workflow: turning application needs into measurable system objectives and evaluation protocols.

Education

- **Donghua University** *2022 - Present*
  Ph.D. Candidate, Software Engineering Shanghai, China
  - Research focus: IoT security, mobile sensing, and computing
- **Jiangsu University of Science and Technology** *2017 - 2020*
  Master, Electronic and Information Engineering Zhenjiang, China
  - Research focus: computer vision, microservices, and modern communication technology
- **Tianjin Renai College** *2013 - 2017*
  Bachelor, Electronic and Information Engineering Tianjin, China
  - Major courses: Principles of Communication, Signals and Processing, Digital Signal Processing, Digital Circuits, Analog Circuits, Principles of Microcomputer and Interface Technology, Principles and Techniques of Single-Chip Microcomputer, C Programming, FPGA Technology

Projects and Grants

- **Doctoral Student Innovation Fund** *Jul 2025 – Jul 2026*
  Project Lead Donghua University
  - Project: Perception Enhancement for Wearable Devices with MEMS Sensors
  - Research directions: IoT, novel sensing networks, ubiquitous computing
  - Sponsor: the Fundamental Reasearch Fund for the Central Universities
  - Grant No: CUSF-DH-D-2025031

Publications

C=Conference, J=Journal, S=Submitted

[C.1] **Luo Zhou**, Shan Chang, et al. "Baro2Talk: Reconstructing Spectrograms from Ear Canal Pressure for Voice-free Communication," in *Proceedings of IEEE International Conference on Computer Communication (INFOCOM)*, 2026. (Accepted)

[C.2] **Luo Zhou**, Shan Chang, et al. "BaroAuth: Harnessing Ear Canal Deformation for Speaking User Authentication on Earbuds," in *Proceedings of IEEE International Conference on Distributed Computing Systems (ICDCS)*, pp. 111–121, 2025. (Glasgow, UK, oral prsentation)

[S.1] **Luo Zhou**, Shan Chang, et al. "Speak and Be Known: Authenticating Users via Ear Canal Deformation on Earbuds," *IEEE Transactions on Mobile Computing (TMC)*. (minor revision: Under Review-Revision 2)

[S.2] **Luo Zhou**, Huixiang Wen, et al. "AirSpy: An Air-pinch Keystroke Inference Attack on Virtual Keyboards in VR Systems," *IEEE Transactions on Consumer Electronics (TCE)*. (minor revision: Under Review-Revision 1)

[J.1] **Luo Zhou**, Xueying Gong, et al. "Mining Interaction-Intensive Cores in Temporal Graphs," *Journal of King Saud University Computer and Information Sciences (JSCI)*, 38(22), 2026.

[C.3] **Luo Zhou**, Yan Xia, and Zhen Liu. "Super-Resolution Reconstruction of Remote Sensing Images Based on GAN," in *Proceedings of IEEE International Conference on Data Science and Computer Application (ICDSCA)*, pp. 270–274, 2021. (DALIAN, China, poster prsentation)

[J.2] Shan Chang, **Luo Zhou**, et al. "Combating Voice Spoofing Attacks on Wearable via Speech Movement Sequences," *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 22(1), pp. 819–832, 2024. (advisor, first author)

[C.4] Jiusong Luo, Shan Chang, and **Luo Zhou**, et al. "Aclipse: Attention-Based Cascaded Learning Enabling Privacy-Preserving Speech Emotion Recognition," in *Proceedings of IEEE International Conference on Parallel and Distributed Systems (ICPADS)*, pp. 1–8, 2025. (Hefei, China, oral prsentation)

[S.3] Jiusong Luo, Shan Chang, and **Luo Zhou**, et al. "Two-Stage Attention-Guided Perturbation for Privacy-Preserving Speech Emotion Recognition," *ACM Transactions on Privacy and Security (TOPS)*. (Under Review)

[C.5] Huixiang Wen, Shan Chang, and **Luo Zhou**. "Light Projection-based Physical-World Vanishing Attack Against Car Detection," in *Proceedings of IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, 2023. (RHODES, GREECE, poster prsentation)

[J.3] Huixiang Wen, Shan Chang, and **Luo Zhou**. "OptiCloak: Blinding vision-based autonomous driving systems through adversarial optical projection," *IEEE Internet of Things Journal (IOT-J)*, 11(17), pp. 28931–28944, 2024.

[J.4] Guodong Wang, Dongyang Ma, **Luo Zhou**, et al. "A Secure Multimodal Retrieval Framework for Consumer Electronics under AI-driven Attacks," *IEEE Transactions on Consumer Electronics (TCE)*, 2025. (Early Access)

[C.6] Shizong Yan, Huixiang Wen, Shan Chang, Hongzi Zhu, and **Luo Zhou**. "Fooling 3D Face Recognition with One Single 2D Image," in *Proceedings of ACM International Conference on Multimedia (MM)*, pp. 4043–4052, 2024. (Melbourne, Australia, oral prsentation)

[J.5] Shizong Yan, Huixiang Wen, Shan Chang, Hongzi Zhu, and **Luo Zhou**. "Closed-Box 3-D Face Reconstruction Attack on Face Recognition From a Single Image," *IEEE Internet of Things Journal*, 12(24), pp. 52179–52193, 2025.

PATENTS AND INTELLECTUAL PROPERTY

[P.1] **Luo Zhou**, Shan Chang, et al. "A user authentication method and system based on ear canal dynamic deformation," 202511388133.5[P], 2025-11-28. (National invention patent)

[P.2] **Luo Zhou**, Shan Chang, et al. "Cross-Modal Silent Speech Reconstruction Method and System Based on Ear-Canal Pressure Micro-Motion Sensing," Patent under application.

[D.1] **Luo Zhou**, Shan Chang. "Ear pressure biometric database driven by human speech activities," Data intellectual property, ID: SZ2025140014216.6.